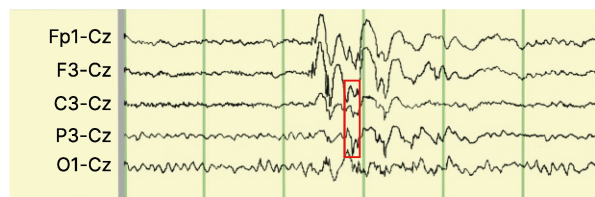


• JUVENILE MYOCLONIC EPILEPSY

- JME syndrome
- Idiopathic > polygenetic
- 10-19 years old
- Myoclonic jerks
- EEG: 4-6 Hz polyspike pattern
- Poor prognosis → develops GTCS in future
- T/t: Valproate



4-6 Hz Polyspike pattern

Important Information

Causes of Myoclonic jerks

< 1 year: infantile spasms

8 years: SSPE

10 - 19 years: JME

30 years: Beef tainted: Startle myoclonus + dementia vCJD

• MTLE: MESIAL TEMPORAL LOBE EPILEPSY

- MRI: Hippocampal sclerosis
- History of febrile convulsion in childhood
- Automatism
- Focal seizures with impaired awareness
- EEG shows U/L or B/L temporal spikes
- DOC: Valproate
- T/t of choice: Neurosurgery

GTCS	FOCAL SEIZURE (mnemonic: L2 COP)	TYPICAL ABSENCE	ATYPICAL ABSENCE ATONIC MYOCLONIC SEIZURES
• Valproate • Women in reproductive age: levetiracetam, lamotrigine (should be given before conception)	• Carbamazepine: anemia, leucopenia, hepatotoxic • Levetiracetam: elderly patients (to avoid interaction other drugs) • Lamotrigine: SJS risk • Oxcarbazepine: safe in liver diseases • Phenytoin	Valproate If planning for pregnancy give Lamotrigine before conception	Valproate If planning for pregnancy give Lamotrigine (before conception)

FEBRILE SEIZURES

- Prevention of episode of febrile seizure: Oral Clobazam > oral diazepam
- During an episode: Intranasal midazolam > rectal diazepam

01:29:52

INFANTILE SPASM

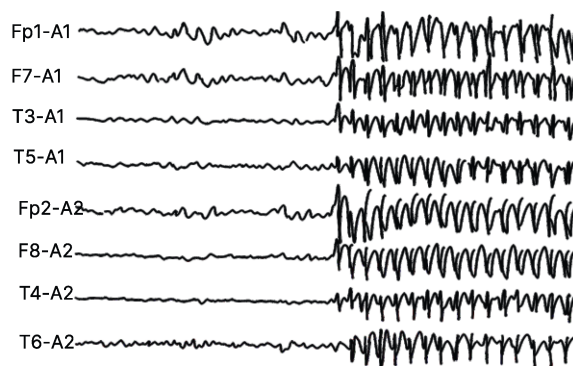
- DCC: AETH
- DCC with Tuberous sclerosis: Vigabatrin

POINTERS:

- M/c type of epilepsy in children: **Benign Rolandic epilepsy**
 - Twitching on one side of the face
 - Involuntary movements on the hand
- M/c type of seizure in children: Febrile seizure
- M/c type of seizure in newborns: Subtle seizure
 - Leading cause of Subtle seizure → hypoxic ischemic encephalopathy (birth asphyxia)

EKG PATTERNS	CONDITION
3 Hz spike and slow wave	Absence seizure
2.5 Hz spike and slow wave	Atypical absence seizure
4-6 Hz polyspike	JME/ JANZ syndrome
Periodic sharp wave complexes (PSCW)	vCJD
Periodic lateralized epileptiform discharges (PLED)	HSV-1 encephalitis
Burst suppression pattern	Cortical anoxia
Triphasic waves	Hepatic encephalopathy vCJD

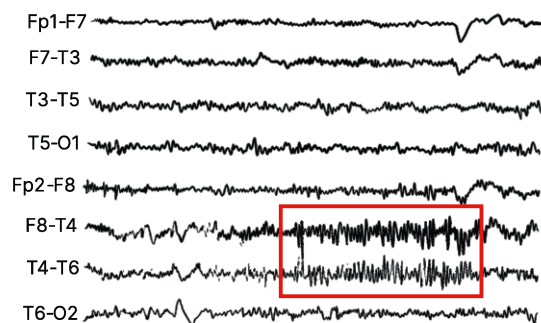
ABSENCE SEIZURES:



- Sudden ↑ in the amplitude of voltage
- **3/per spike pattern**

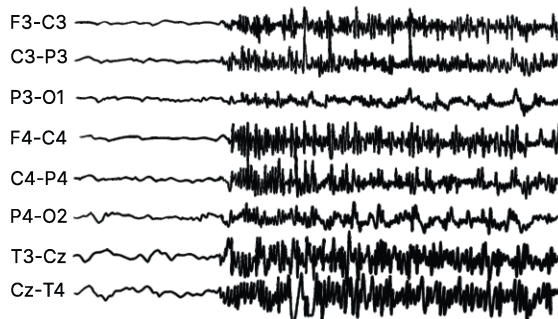
FOCAL SEIZURE

- Epileptic form discharge at F8-T4, T4-T6
- Rest showing background rhythm

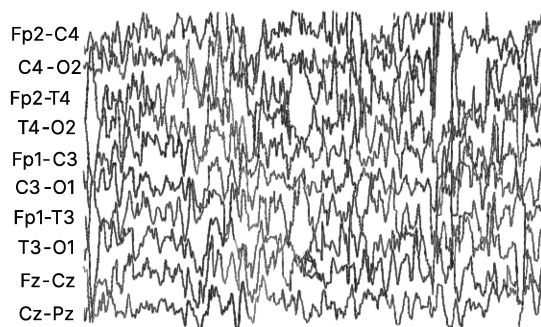


DIFFUSE GTCS

- High amplified epileptic form discharges in all leads



HYPERSARRHYTHMIA

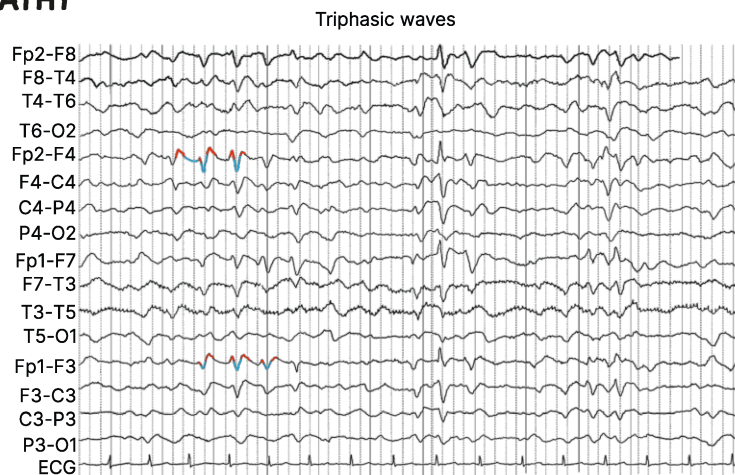


- No fixed pattern
- Very high amplitude complexes with low frequency
- Chaotic pattern
- Seen in infantile spasm

VCJD



- Periodic sharp wave complexes



- Negative (red) and positive (blue) deflections in Fp2-F4 and Fp1-F3
- Positive deflection preceded and followed by negative deflection → **Triphasic waves**
- Triphasic waves seen in hepatic encephalopathy and vCJD

SSPE: SUBACUTE SCLEROSING PAN ENCEPHALITIS

01:39:59

- Late neurological **complication of measles**
- Late neurodegeneration condition, when the infection is of altered measles virus
- Low vaccine coverage area
- Infection at 2 years of age → clinical manifestations at 8 years of age → **slow virus disease**

Clinical features	Progressive, subacute mental deterioration with typical signs like myoclonus
EEG	Periodic, stereotyped, high-voltage discharges
CSF	Raised gamma globulin or oligoclonal pattern
Measles antibodies	Raised titre in serum (1:256) and/ or cerebrospinal fluid (1:4)
Brain biopsy	Suggestive of panencephalitis

Definitive: criteria 5 with three more criteria, three of the five criteria

PHAKOMATOSIS/ NEUROCUTANEOUS DISORDER (TUBEROUS SCLEROSIS)

01:45:08

- Genetics
 - Autosomal dominant
 - TSC1 gene: chromosome 9
 - TSC2 gene: chromosome 16
 } Defect in any of these genes
- Cutaneous manifestations
 - Ash leaf macule (hypopigmented lesion)
 - Shagreen patch (orange peel consistency)
 - Nail: Koenen tumor/ subungual fibroma
 - Acneiform lesion seen in puberty adenoma sebaceum



- Neurofibroma - potato like lesions → subependymal astrocytomas
- Kidney: angiomyolipoma
- Angioma + bleeding → Wunderlich syndrome



Adenoma Sebaceum



Ash leaf macule



Shagreen patch



Subungual fibroma

STURGE WEBER SYNDROME

- Port wine stain on the face + CNS manifestations
- Features: **STURGE**
 - **S** - seizures (cavernous angioma)
 - **T** - trigeminal nerve distribution - port wine stain (vascular malformation)
 - **U** - U/L weakness
 - **R** - mental retardation
 - **G** - glaucoma
 - **E** - eye: **buphthalmos**
- CT Head: visible calcification in the brain - **Tram track appearance**



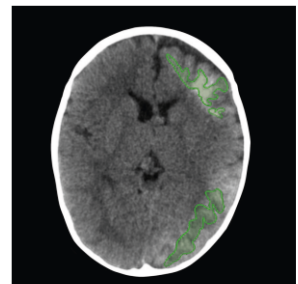
Port wine stain

Important Information

Tram track appearance

HRCT - Chest: Bronchiectasis

Kidney biopsy: MPGN



Tram track appearance



Match the given waves with their appropriate frequency and the state of non-rapid eye movement sleep they are observed in:

1. Delta waves	a) 4-7 Hz	i) Stage 1
2. Mu waves	b) 12-14 Hz	ii) Stage 3
3. Theta waves	c) 0.5-4 Hz	iii) Stage 2

- a. 1-b-ii, 2-c-i, 3-a-iii
- b. 1-a-i, 2-c-iii, 3-b-ii
- c. 1-c-iii, 2-b-ii, 3-a-i
- d. 1-c-ii, 2-b-iii, 3-a-i

Ans: (c)

Which of the following statements is true regarding myoclonus?

- 1. Physiological myoclonus is the sudden jerking movement that may occur when person fall asleep
- 2. Startle myoclonus with dementia is seen in vCJD
- 3. Drug of choice is Valproate
- 4. GTCS is seen in cases of Juvenile myoclonic epilepsy later

- a. 1 and 3
- b. 2 and 4
- c. 1, 2 and 4
- d. 1, 2, 3 and 4

Ans: (d)

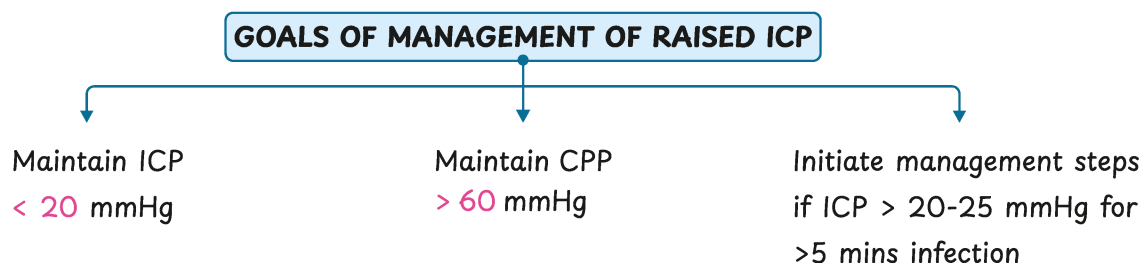


RAISED ICP AND BRAIN DEATH

RAISED INTRACRANIAL PRESSURE

00:00:31

- **Normal** ICP - 10 - 20 mm Hg/50 - 180 mm H₂O
- Raised intracranial pressure: > 25 mmHg
- Cerebral perfusion pressure (CPP): Mean arterial pressure - Intracranial pressure
 - Mean arterial pressure = Systolic BP + 2 (Diastolic BP) divided by 3
 - CPP = 95 mmHg - 20 mmHg = 75 mmHg
 - Range of **CPP** - 60 to 80 mmHg
 - **CPP < 60mmHg** → brain infarction



BRAIN HERNIATION

00:05:09

TYPES

1. TRANSTENTORIAL HERNIATION

- M/c herniation
- **Inferior herniation** of brain

2. SUBFALCINE HERNIATION

- Midline shift
- **Horizontal displacement**

3. UNCAL HERNIATION

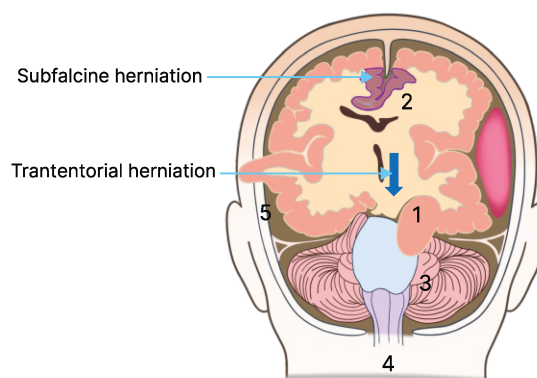
- Compression of CN III
- I/L mid dilated pupil (**Hutchinson pupil**)
- Sluggish reaction to light

4. FORAMINAL HERNIATION

- **Inferior displacement of cerebellar tonsils**
- D/t LP procedure in patient with raised ICP
- Compression of medulla → apnea

5. TRANSCALVARIAL HERNIATION

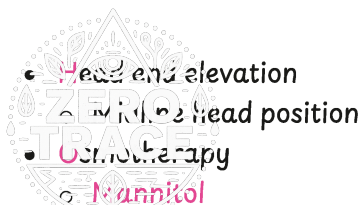
- Brain tissues **displaced into skull** after trauma
- Bad prognosis



STEPS IN MANAGEMENT OF RAISED ICP

00:10:35

- Insert ICP monitor - ventriculostomy
 - **Highly effective way to manage raised ICP**



- Head and elevation
 - Maintain head position
- Osmotherapy
 - Mannitol
 - C/I - active CNS bleeding (warfarin toxicity)
 - 23.4% hypertonic saline
 - 5% dextrose absolutely C/I (Hypotonic solution → worsens condition)
- Drain CSF
 - Via ventriculostomy (not possible in most of settings in India)
- Glucocorticoids
 - Vasogenic cerebral edema → damaged BBB (accumulation of water in interstitial spaces)
 - Seen with Tubercular meningitis, NCC, Brain tumor, HACE
 - Not effective in stroke, traumatic brain injury (cytotoxic cerebral edema)
- Sedation
 - Midazolam, Propofol, Morphine infusion
 - Neuromuscular paralysis – intubation, PPV
 - minimises risk of barotrauma and maintains patient on ventilator
- Hyperventilation (short term step)
 - ↑ respiratory rate → CO_2 washout → vasoconstriction in brain → ↓ CSF production → preventing further ↑ of ICP
- Pressor therapy
 - Helps in ↑ MAP
 - Dopamine, Norepinephrine, Phenylephrine
- Teir II therapy for refractory raised ICP
 - Decompressive hemicraniectomy
 - Pentobarbital coma
 - Barbiturate - cerebroprotective → ↓ metabolism → ↓ energy requirement
 - Therapeutic hypothermia (33°C) → ↓ metabolism

VENTRICULOSTOMY

00:28:27

- Most effective method
- Procedure

Drill hole at top of head
↓
catheter is placed in lateral ventricle (ICP monitoring catheter)
↓
sensor that is attached at tip of catheter will record ICP >25 mmHg

DECOMPRESSIVE HEMICRANIECTOMY

Removal of bone flap
↓
Incision of dura to release pressure

Important Information

Physiology of CSF

- Site of production: Choroid plexus
- Site of reabsorption: Arachnoid granulation along the venous system of superior sagittal sinus
 - Damage to arachnoid granulation leads to ↓ reabsorption of CSF
- Total volume of CSF: **150 ml**
- Rate of production: **20 ml/hr**
- Rate of reabsorption: **20 ml/hr**
- ICP: **10-20 mmHg**
- When compared to plasma, CSF has higher sodium, chloride and magnesium but lower calcium and potassium

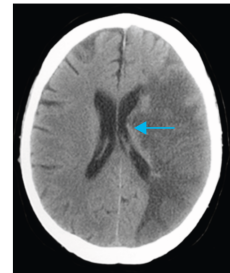
VASOGENIC CEREBRAL EDEMA

- Occurs due to ↑ permeability of BBB
- Causes ↑ interstitial fluid between neurons
- Affects white matter and grey matter **spared**
- E.g. Tubercular meningitis - ATT + Steroids



CYTOTOXIC CEREBRAL EDEMA

- Occurs due to stroke
- Causes intracellular fluid accumulation
- **Affects** both grey and white matter
- Neuroimaging shows -
 - Loss of grey-white matter differentiation



CLINICAL FEATURES OF RAISED ICP

00:35:27

SYMPTOMS

- Headache
 - **Frontal** (initially) d/t stretching of dura mater (pain sensitive structure), later holocephalic
 - CN V nociceptive receptor gets stimulus
- Projectile vomiting - Vomiting **without** nausea
- Diplopia on looking laterally (**false localizing sign**) - Stretching of CN VI
- Altered mental status
 - Most reliable symptom
 - Reticular activating system - responsible for alertness and being awake gets damaged

Important Information

- CN with longest subarachnoid course - **CN VI**
- M/c CN involved in raised ICP: CN VI
- False localizing sign: CN VI
- CN with longest intracranial course: **CN IV**
- Overall longest CN: CN X (Vagabond nerve)
- CN with longest intraosseous course: **CN VII**



EXAMINATION FINDINGS

• **Heart Rate** bradycardia

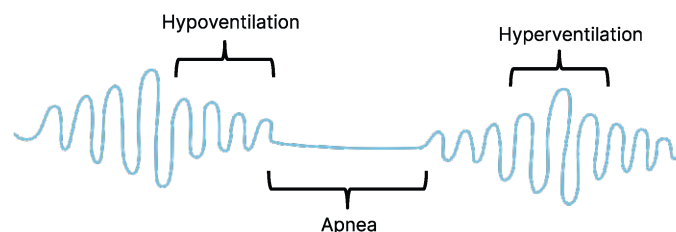
• **BP** 1

• **Breathing**

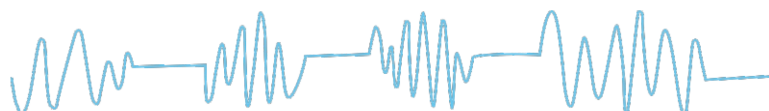
○ Cheyne Stokes breathing/Periodic breathing

○ Seen in - ↑ ICP, CHF, HAPE

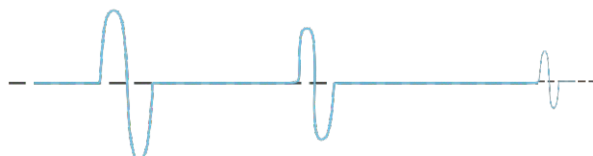
Cushing reflex } **Cushing triad**



○ Biot's breathing - irregularly irregular breathing



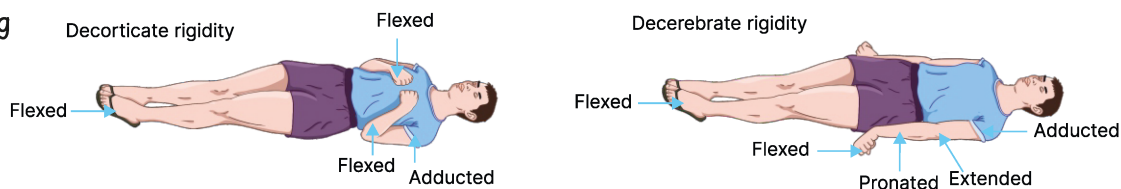
○ Agonal rhythm



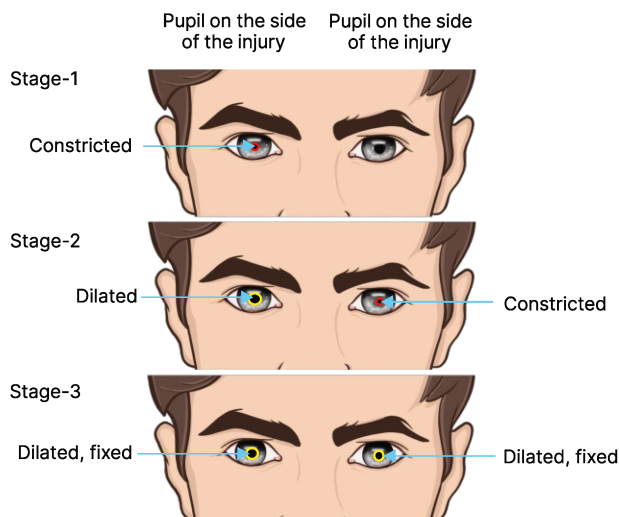
Important Information

- Irregularly irregular breathing: **Biot's breathing**
- Irregularly irregular pulse: **Atrial fibrillation**

• **Posturing**



• **Hutchinson's pupil** - I/L mid dilated pupil, sluggish reaction to light

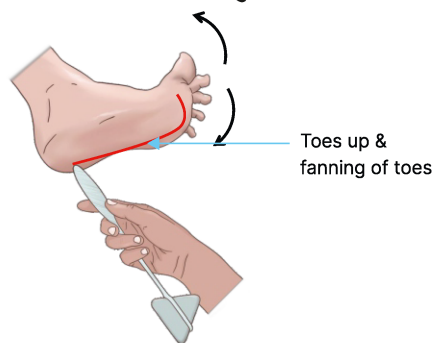


Free By - ZeroTrace

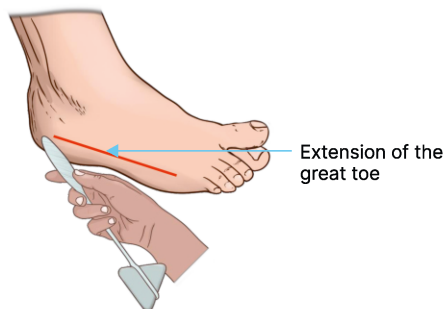


• Babinski DTR

- Stimulation of corticospinal pathway → loss in inhibitory control → brisk reflex
- **Babinski sign** - stimulating sole of foot → dorsiflexion of great toe, fanning of toes



- **Chaddock sign** - stimulation at lateral aspect of foot towards small toe → extension of great toe, fanning of toes



- **Gordon sign** - squeezing of calf muscle → extension of great toe, fanning of toes



- Fundus examination - **papilledema**

Important Information

- Rubrospinal tract causes excitation of limb flexors and inhibits antagonist extensors
- Decerebrate posturing: Rubrospinal tract damaged, causes loss of flexor tone and extension of upper limbs seen
- Decorticate posturing - lesion above the red nucleus, Upper mid brain
 - Loss of inhibitory control over rubrospinal pathway → flexion of upper limbs
- Vestibulospinal and Reticulospinal tract mediates extension of legs

- Anisocoria, pupils, reactive to light -
- Metabolic encephalopathy - Fulminant hepatic failure (Ammonia intoxication), AKI (↑ urea)
- Pinpoint pupils

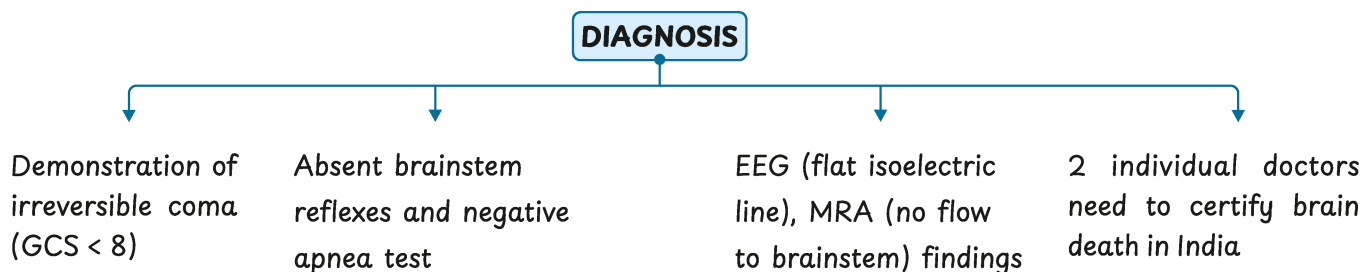
FEATURES	TEMPERATURE	RESPIRATORY RATE
Pontine stroke	↑	↑ (irregular respiration)
Heroin overdose	↓	↓↓

Cortical stroke	• Patient looks towards side of lesion (looks away from side of weakness)
Brainstem stroke	• Patient looks towards side of lesion • Crossed feature • I/L CN palsy + C/L hemiplegia
Ocular dipping	• Diffuse cortical anoxia (Cardiorespiratory arrest) • Slow downward and fast upward movement of eyes
Ocular bobbing	• Basilar artery involvement • Pontine stroke • Fast downward and slow upward movement of eyes

BRAIN DEATH

01:00:13

- Brain stem death, Heart still beating
- Twitching toes/ fingers, DTR elicitable



CAUSES OF REVERSIBLE COMA

- Diabetes ketoacidosis, Hyperosmolar coma, Uremic encephalopathy (T/t with hemodialysis)
- Fulminant hepatic failure, hepatic encephalopathy, electrolyte imbalance , drug overdosage,

DIAGNOSIS

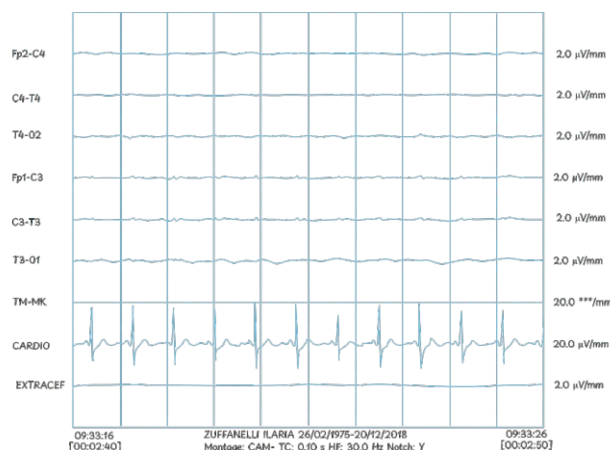
- Random blood sugar, urine for ketones, drugs
- Kidney function tests
- Liver function tests
- S. electrolytes

REFLEXES	INTERPRETATION	CN TESTED
Light reflex	Pupils- fixed, non-reactive to light	II, III (mid brain)
Corneal reflex	Blinking of eyes - B/L absent	V, VII (pons)
Cough reflex	Cough - suction catheter	X, X (medulla)
Caloric stimulation test	C. O. W. S - negative Cold water - opposite side nystagmus Warm water - same side nystagmus	VIII, III, VI
Oculocephalic test	Doll's eye reflex - negative	VIII, III, VI
Gag reflex	Gag and cough	IX, X

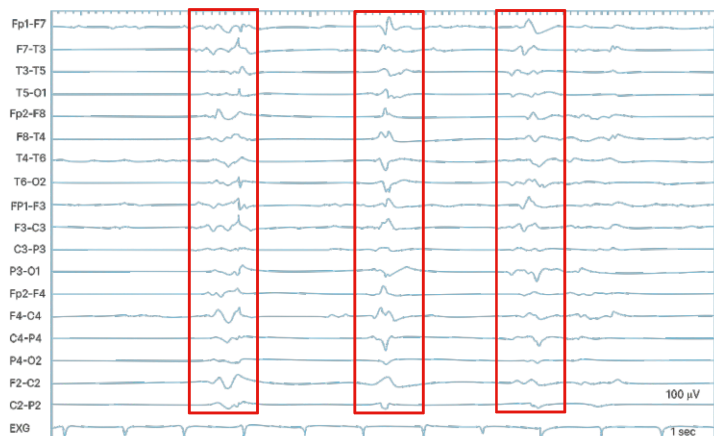
APNEA TEST

- Preoxygenate with 100% O₂ for 10 mins so that pO₂ becomes 200 mmHg then disconnect ventilator
- pCO₂ > 60 mmHg (most potent stimulus at respiratory center)
- Absence of spontaneous breathing efforts
- Pacemaker of respiratory center - **Pre-Botzinger** complex → completely damaged

EEG



Flat isoelectric line



Burst suppression pattern

- ↑ Sensitivity - 2 µV
- Burst suppression pattern initially - seen with **cortical anorexia**
- Flat isoelectric line >30 min

ORGAN TRANSPLANTATION

01:15:44

- Brain dead person can save 8 lives

ORGAN TRANSPLANTATION	CAUSES
Heart	Cardiomyopathy
Lungs	Interstitial lung disease
Liver	Nonalcoholic fatty liver disease
Kidney	Diabetic nephropathy
Pancreas	Type I DM
Small intestine	Short gut syndrome



Q. Which of the following is not used in management of traumatic brain injury?

- a. Dexamethasone
- b. Hypertonic saline
- c. Hyperventilation
- d. External ventricular drainage

Ans: (a)

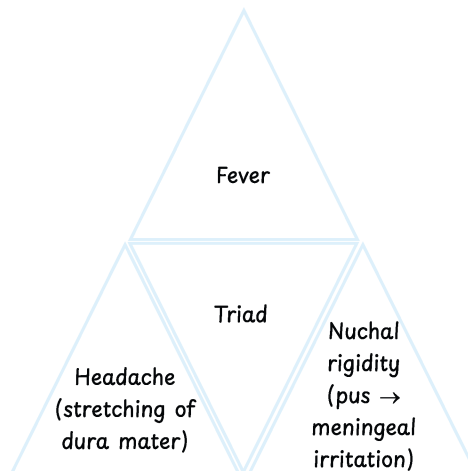


MENINGITIS PART 1



MENINGITIS

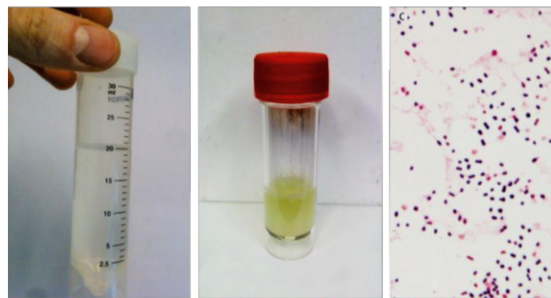
00:00:10



- Triad - present in 80% of cases
- Cases of late diagnosis - altered mentation, 6th nerve palsy (*false localizing sign*)

BACTERIAL MENINGITIS

00:01:47



Turbid CSF with neutrophils

NEONATAL MENINGITIS

- *M/c* cause - *Streptococcus agalactiae*/ Group B *Streptococcus* (globally)
- *M/c in India* - *E. coli*/ *Klebsiella*
- Antibiotic given for these infections - Cefotaxime
- Rare cause - *Listeria monocytogenes*
 - Resistant to Cefotaxime
 - Sensitive to Ampicillin
 - *Palcam media* → CSF culture

CHOICE OF EMPIRICAL ANTIBIOTICS IN PYOGENIC/ BACTERIAL MENINGITIS

	Antibiotics
Neonates (preterm to term neonates up to 3 months)	Ampicillin + Cefotaxime (3 rd gen cephalosporin)
Infants > 3 months to adults < 55 years	Ceftriaxone + Vancomycin (added d/t possibility of ceftriaxone resistance)
Adults >55 years/ Alcoholics/ debilitating illness	Ampicillin + Cefotaxime
Neutropenic patient, post neurosurgery, post trauma hospital acquired meningitis	Ampicillin + Ceftazidime or Meropenem (in cases of Ceftazidime resistance)

MENINGITIS IN CHILDREN

00:07:09

- M/c organism - *Pneumococcus* > *H. influenzae* (d/t pentavalent vaccine)

MENINGITIS IN ADULTS

- Leading causes in adults -
 - M/c - *Pneumococcus* (Sporadic)
 - *N. meningitidis* (Outbreaks in winter)
 - Spread via **nasal droplets**
 - Vulnerable population - Prison/ Hostel/ Slums
 - Meningitis
 - Sepsis : DIC → ↓ platelets → petechiae/rash → adrenal hemorrhage
 - Adrenal hemorrhage → ↓ BP; ↓ sugar levels → Addisonian crisis
- } **Waterhouse Friderichsen syndrome**



Bilateral adrenal hemorrhage



Petechial hemorrhages

CHEMOPROPHYLAXIS FOR MENINGOCOCCAL MENINGITIS

00:12:12

- Health care worker - *Ceftriaxone injection* (I/M)
- Family contact - *Ciprofloxacin* (oral)
- Pregnancy - *Ceftriaxone injection* (I/M)
 - *Ciprofloxacin* in 1st trimester → potential teratogenesis
 - Quinolones → impaired cartilage growth in children



Important Information

- **Reynolds syndrome**
- Internal bleeding in kidneys
- Seen in Tuberous sclerosis - Angiomyolipoma

VIRAL MENINGITIS

00:16:22

VIRAL MENINGITIS	VIRAL ENCEPHALITIS
• (+) Nuchal rigidity • Cause : Enterovirus	• (-) Nuchal rigidity • Global - HSV-1 encephalitis ◦ Temporal lobe damage predilection ◦ T/t → empirical Acyclovir • India ◦ Japanese B encephalitis → Transmission - Culex mosquito → No cure (only vaccines) ◦ Nipah virus (Kerala) • Transmission - fruit bat (Pteropus)

HSV-1 ENCEPHALITIS

00:20:22

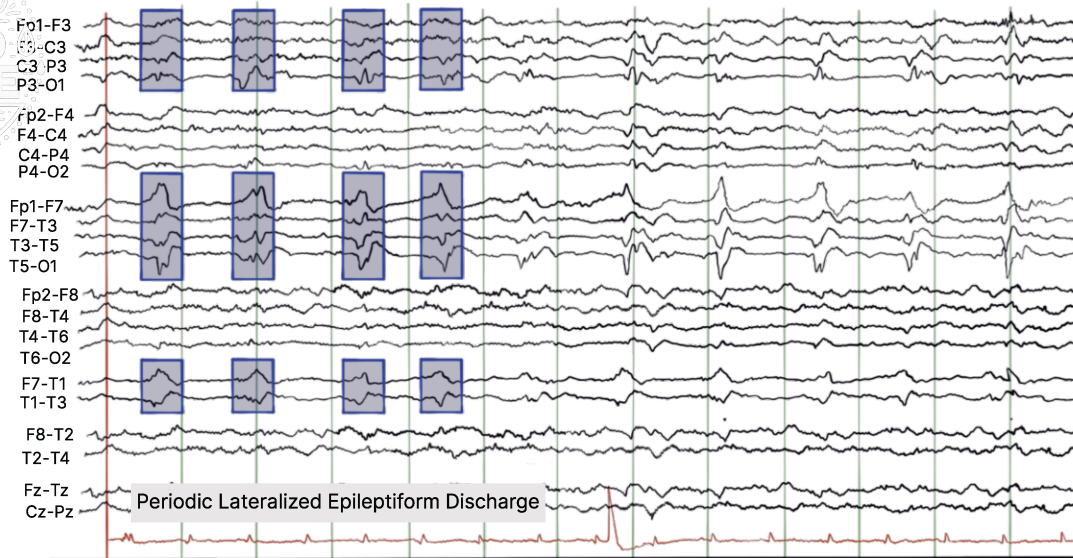
- CSF findings - Bloody CSF
 - Bloody CSF seen in - SAH, Traumatic LP, HSV-1 encephalitis
- **EEG** finding - Periodic lateralized epileptiform discharge
 - Periodic Sharp Wave Complex → Variant CJD
- **IOC** - PCR of CSF for HSV-1

CASE SCENARIO

Q. Young man presents with high grade fever for 1 day with violent psychotic behavior, mutism and not recognizing family members. GCS- 9/15 and bilateral Babinski sign is noted

Differential diagnosis :

- **HSV-1 encephalitis**
- If lab shows evidence of hemolysis, elevated LDH, ↑ unconjugated bilirubin and peripheral smear shows multiple ring stages in RBC
 - **Cerebral malaria**
- Both diagnosis have diffuse cerebral malfunctions



SCENARIO-1

- Grouped vesicles on glans penis/ vulva
- HSV-2



Group vesicles on glans penis/vulva

SCENARIO-2

- Grouped vesicles in dermatomal distribution
- Herpes zoster



Group vesicles in dermatomal distribution

Q. An 18-year-old went back to his village for holidays and used to swim daily in a village pond. After a week he developed fever, frontal headache with projectile vomiting and nuchal rigidity. LP was done and CSF shows 10,000 PMN cells with motile trophozoites. What is the likely diagnosis?

Ans - Primary amoebic meningoencephalitis

- Organism responsible - *Naegleria fowleri*
- Route of entry - Olfactory neuroepithelium
- T/t: Amphotericin B, Miltefosine

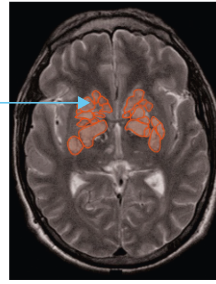
FUNGAL MENINGITIS

- Leading cause - *Cryptococcus neoformans* (inhalational dust from pigeon droppings)
- IOC: CSF ELISA for cryptococcal antigens
- T/t: Liposomal Amphotericin B + 5 flucytosine
 - Prophylaxis: Fluconazole (until CD₄ count ↑)

00:28:56



Soap bubble appearance in Cryptococcosis



MRI

Important Information

Lyme disease	Caused by <i>Borrelia</i> • Manifestations: ◦ Lymphocytic meningitis ◦ Cranial neuritis (CN-VII is usually involved) ◦ Radiculoneuritis • T/t : Doxycycline/ Ceftriaxone
CSF shunts meningitis	• CONS, <i>S. aureus</i>
Aseptic meningitis	• No organism identified (Later may be reclassified to viral meningitis)
Complications of Meningitis	• Abscess, cerebritis, thrombosis • Hydrocephalus • SIADH → Hyponatremia